

Geometry-Aware Scene Configurations for Novel View Synthesis

Minkwan Kim*

Changwoon Choi†

Young Min Kim‡

Seoul National University

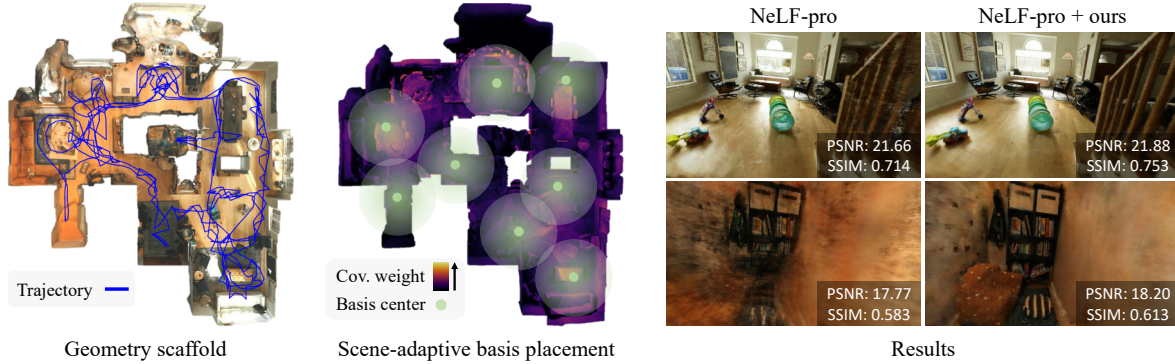


Figure 1: Geometry-aware basis configuration and novel view synthesis results. After recording measurement statistics as coverage weights on geometry scaffold, we find optimal basis positions and training-view configurations in indoor environments.

ABSTRACT

We introduce scene-adaptive strategies for efficiently allocating representation capacity to enable immersive indoor scene rendering. We maximize the use of limited resources, leveraging geometric priors that are typically available after pre-processing. By recording observation statistics on an estimated geometric scaffold, we guide adaptive basis placement, significantly enhancing uniform arrangements used in prior scalable scene representations. We further propose scene-adaptive virtual viewpoints to address geometric deficiencies in input view trajectories and provide effective regularization. Our approach yields substantial improvements in rendering quality over methods based on regular basis placement.

Index Terms: Neural Radiance Fields, free view synthesis, indoor scene reconstruction.

1 INTRODUCTION

Recent advances in Neural Radiance Fields (NeRF) [4] have enabled immersive 3D scene exploration. However, existing NeRF-based methods [5] struggle to scale to large scenes due to limited representation capacity and inefficiencies in handling complex scenes. Moreover, data captured in everyday environments often suffer from inaccurate geometry and non-uniform ray measurements, leading to degraded performance compared to forward-facing or object-centric evaluation settings. To address scalability, recent works [9] employ multiple bases—spatial entities like NeRF blocks or feature grids—to represent unbounded scenes. However, these methods typically rely on simple heuristics, such as placing bases uniformly [6] or along camera trajectories [9] without considering complex scene layout and observation statistics.

We propose a systematic approach that leverages the interaction between scene geometry and camera viewpoints. By utilizing

geometric priors which are readily available, we introduce scene-adaptive strategies for efficient resource utilization. We optimize basis placement that improves scalability and reconstruction quality by analyzing the correlation between scene geometry and input observations. Our method further accounts for insufficient observations to enhance novel view synthesis. We identify virtual viewpoints that compensate for missing measurements using geometric structures and impose depth regularization to reduce artifacts. These strategies significantly improve rendering quality while minimizing memory usage, as validated on the ScanNet++ [8] dataset.

2 METHOD

Given free-trajectory images of large-scale indoor scenes, we obtain geometric scaffolds from recent neural implicit surface reconstruction methods [10]. We then use these geometric scaffolds to optimize basis placement, where each basis comprises local scene representation. Based on the geometry and observation statistics, we further learn the radiance volume using both the input trajectory and virtual viewpoints under depth supervision.

2.1 Basis Placement Optimization

We propose an adaptive approach that optimizes the distribution of bases by leveraging scene geometry and camera trajectories. We summarize observation statistics on the geometric structure using a coverage weight w_i , which is computed for a surface point $x_i \in \mathbb{R}^3$ on the mesh as follows:

$$w_i = \sum_k \frac{\hat{n}_i \cdot (c_k - x_i)}{\|c_k - x_i\|^3} \text{Vis}(c_k \rightarrow x_i), \quad (1)$$

where $\hat{n}_i$ denotes the unit surface normal at x_i , c_k is the camera center, and $\text{Vis}(\cdot)$ indicates binary visibility. This weight accumulates observation frequency by accounting for visibility, distance, and surface orientation.

The coverage loss $\mathcal{L}_{\text{cov}}$ encourages bases to move toward frequently observed regions by penalizing the distance between basis locations $p_j \in \mathbb{R}^3$ and surface points x_i :

$$\mathcal{L}_{\text{cov}} = \sum_i w_i \left(\frac{\|p_{C(i)} - x_i\|^3}{\hat{n}_i \cdot (p_{C(i)} - x_i) + \epsilon} \right), \quad (2)$$

*e-mail: mkjjang3598@snu.ac.kr

†e-mail: changwoon.choi00@gmail.com

‡e-mail: youngmin.kim@snu.ac.kr

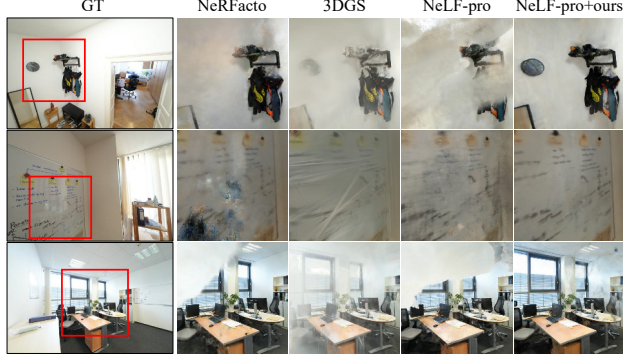


Figure 2: Qualitative results of novel view synthesis on ScanNet++ [8] dataset. Best viewed on screen.

Table 1: Performance comparison on ScanNet++ [8] dataset.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Memory	Time
MonoSDF [10]	20.41	0.800	0.351	61 MB	6 h
NeRFacto [5]	20.52	0.793	0.352	65 MB	14 min
Depth-NeRFacto [5]	21.75	<u>0.808</u>	0.333	65 MB	17 min
3DGS [1]	21.26	0.797	0.305	164 MB	12 min
Hierarchical-3DGS [2]	20.57	0.771	0.335	793 MB	41 min
NeLF-pro [9]	21.88	0.795	0.367	89 MB	21 min
NeLF-pro [9] + Ours	22.52	0.814	<u>0.329</u>	89 MB	41 min

where $C(i)$ denotes the index of the basis assigned to surface point x_i and ε is a small constant for numerical stability. The optimized basis locations p_j are used as spatial units that constitute the global radiance fields, implemented either as centers of NeRF blocks [3] or as neural light field probes [9].

2.2 Scene-Adaptive Geometric Regularization

To reduce floating artifacts under extreme viewpoint changes, we apply geometric regularization using the geometric scaffold. Since the input camera trajectory often fails to sufficiently cover all regions of the scene, we introduce depth maps from additional viewpoints into a set S to improve coverage. These additional viewpoints are progressively selected by sampling candidate views V from unobserved regions. The selection criterion maximizes the distance to the existing viewpoints while accounting for scene diversity, which is computed using a similarity matrix of observed feature points. Motivated by Xiao et al. [7], the selection rule is defined as:

$$v^* = \arg \max_{v \in V \setminus S} \left[\min_{s \in S} d(v, s) \right], \quad (3)$$

where the viewpoint distance $d(v, s)$ is defined as:

$$d(v, s) = \|c_v - c_s\|^2 + \kappa \left(1 - \frac{A_{vs}}{\max(A_{vs})} \right), \quad (4)$$

Here, the first term accounts for spatial expansion by measuring the distance between the candidate viewpoint location c_v and an existing camera position c_s . The second term encourages scene diversity using the similarity matrix A_{vs} , where A_{vs} denotes the number of commonly observed feature points between viewpoints v and s . During training of the final radiance field, the overall loss combines the RGB reconstruction loss, the depth loss for both training and sampled virtual viewpoints, and additional regularization terms in the original NeLF-pro [9].

3 RESULTS & DISCUSSION

We evaluate our framework on the novel view synthesis task using ScanNet++ [8] dataset, which includes high-quality training and

test images along with camera poses. For evaluation, we select six scenes covering both interpolation and extrapolation scenarios. We compare our approach with several baseline methods, including 1) single-basis architectures–MonoSDF [10], NeRFacto [5], Depth-NeRFacto and 3DGS [1] and 2) multiple-basis architectures–NeLF-pro [9] and Hierarchical-3DGS [2].

Following NeLF-Pro, we represent the entire scene with 64 bases and 3 core probes, where 16 nearby basis probes contribute to volume rendering. Our proposed framework significantly outperforms other baselines while maintaining the same number of parameters as NeLF-Pro. Optimized basis placement and geometric regularization enable better reconstructions, particularly in extrapolation scenarios where existing methods (e.g., 3DGS) struggle. Comparative results are summarized in Fig. 2 and Tab. 1. Overall, our adaptive strategy improves coverage compared to uniform or trajectory-based distributions and improves novel view synthesis across the dataset in both interpolation and extrapolation settings. Performance gains are achieved through optimized basis placement and geometric regularization, while memory usage remains identical to the original NeLF-Pro.

4 CONCLUSION

We present scene-adaptive strategies that optimize basis placement by exploiting the correlation between scene geometry and camera trajectories to efficiently utilize limited resources. Leveraging a preprocessed geometric scaffold, we determine adaptive basis placement for large-scale indoor scenes. We further regularize underconstrained geometry using scene-aware virtual viewpoints and geometric priors. Experiments show that our framework consistently outperforms baselines in challenging scenarios, including extrapolation settings. While effective for general indoor environments, our method currently does not model view-dependent reflectance, which may result in artifacts under strong specularities. Extending this framework to in-the-wild scenes with complex lighting and diverse appearance changes remains an interesting direction for future work.

REFERENCES

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), July 2023. 2
- [2] B. Kerbl, A. Meuleman, G. Kopanas, M. Wimmer, A. Lanvin, and G. Drettakis. A hierarchical 3d gaussian representation for real-time rendering of very large datasets. *ACM Transactions on Graphics*, 43(4), July 2024. 2
- [3] A. Meuleman, Y.-L. Liu, C. Gao, J.-B. Huang, C. Kim, M. H. Kim, and J. Kopf. Progressively optimized local radiance fields for robust view synthesis. In *CVPR*, 2023. 2
- [4] B. Mildenhall, P. P. Srinivasan, M. Tancik, J. T. Barron, R. Ramamoorthi, and R. Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *ECCV*, 2020. 1
- [5] M. Tancik, E. Weber, E. Ng, R. Li, B. Yi, J. Kerr, T. Wang, A. Kristoffersen, J. Austin, K. Salahi, A. Ahuja, D. McAllister, and A. Kanazawa. Nerfstudio: A modular framework for neural radiance field development. In *SIGGRAPH*, 2023. 1, 2
- [6] H. Turki, D. Ramanan, and M. Satyanarayanan. Mega-nerf: Scalable construction of large-scale nerfs for virtual fly-throughs. In *CVPR*, 2022. 1
- [7] W. Xiao, R. Santa Cruz, D. Ahmedt-Aristizabal, O. Salvado, C. Fookes, and L. Lebrat. Nerf director: Revisiting view selection in neural volume rendering. In *CVPR*, 2024. 2
- [8] C. Yeshwanth, Y.-C. Liu, M. Nießner, and A. Dai. Scannet++: A high-fidelity dataset of 3d indoor scenes. In *ICCV*, 2023. 1, 2
- [9] Z. You, A. Geiger, and A. Chen. Nelf-pro: Neural light field probes for multi-scale novel view synthesis. In *CVPR*, 2024. 1, 2
- [10] Z. Yu, S. Peng, M. Niemeyer, T. Sattler, and A. Geiger. Monosdf: Exploring monocular geometric cues for neural implicit surface reconstruction. *NeurIPS*, 2022. 1, 2